Chakib OUZANE

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Education

MSc **Master's Degree in Computer Science** — 2020 – 2025
ESI SBA (l'École Supérieure d'Informatique - Sidi Bel Abbes) — Sidi Bel Abbes, Algeria
Specialty: Computer Systems Engineering (ISI)

Eng **State Engineer Degree in Computer Science** — 2020 – 2025
ESI SBA (l'École Supérieure d'Informatique - Sidi Bel Abbes) — Sidi Bel Abbes, Algeria
Specialty: Computer Systems Engineering (ISI)

BAC **Baccalaureate Diploma** — 09/2020
Branch: Mathematics — Sidi Bel Abbes, Algeria

Experiences

System Administrator at GSH (Groupe Sociétés HASNAOUI) — 08/2026 - Present, Sidi Bel Abbès, Algeria

- Managing Active Directory infrastructure, including user accounts, group policies, and access control.
- Supporting broader IT infrastructure operations, including network administration, server maintenance, and troubleshooting.

Internship at CESI LINEACT Research Lab, Machine learning & Network security — 04/2025 - 09/2025, Aix-en-Provence, France

- Comprehensive literature review of Researches on security in IoT Networks.
- Designing, Implementing and Simulating Flooding Attacks on RPL networks using GNS3 & Cooja simulators.
- Comprehensive literature review of Researches on Machine Learning countermeasures for IoT Security.
- Proposing a FL based framework for detecting routing attacks in IoT environments that rely on RPL.

Internship at GSH (Groupe Sociétés HASNAOUI), Network security — 06/2023 - 09/2023, Sidi Bel Abbes, Algeria

- Experimentation with the SOPHOS software.
- Setting SOPHOS in virtual environments, testing filtering policies, Intrusion Prevention (IPS), and deep packet inspection for network traffic protection.

Publications

DRIFT: Distributed RPL Injection Flooding Threat in IoT Networks — 08/06/2026, London, United Kingdom

- Publication at the 14th International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA 2026).
- Conference paper status: Accepted — **Best Paper Award**

TIDE: Threat of Internal DIS Exploit in RPL-based IoT Networks — 2027, Washington DC, USA

- Targeting: IEEE International Conference on Communications (ICC), 2027.
- Conference paper status: In Preparation

Projects

Web-Controlled Autonomous Car Robot with Real-Time Object Detection (Embedded/CPS) 03/2024 - 06/2024
GitHub

- Co-developed a Wi-Fi-controlled robotic vehicle built on ESP32-CAM, with firmware written in C++, integrating motor/servo control and a remote web-based control interface.
- Designed a distributed architecture offloading real-time YOLO object detection from the resource-constrained embedded board to a server, addressing ESP32's limited computing capacity.
- Diagnosed and resolved system-level constraints including power delivery, embedded computing limits, and bandwidth bottlenecks in streaming camera data to a cloud backend (GCP).

Machine Learning-Based Spam Detection in Chicago Hotel and Restaurant Reviews 12/2024

- Built a machine learning pipeline to classify reviews as spam or non-spam using text and metadata features.
- Performed data cleaning, feature engineering (TF-IDF, sentiment scores, reviewer behavior), and model evaluation (Logistic Regression, Random Forest, XGBoost).
- Achieved 89.19% test accuracy with XGBoost; analyzed class imbalance impact and proposed improvements for F1 score.

Supervised Machine Learning for Sales Data Analysis and Prediction

- Performed data preprocessing and feature selection on a supermarket sales dataset.
- Visualized sales trends and spending patterns across branches, customer types, and products using R and ggplot2.
- Prepared dataset splits for training and testing supervised ML models.

Jeunesse Volontaire 2024
jeunessevolontairesba.dz

- A Website + Platform built for an environmental organization.
- Technologies used: SvelteKit & Svelte, Supabase, Sanity CMS, Zod, Tailwind.

Alphabit Club Official Website 2023
www.alphabit.club

- I worked with a team to develop a website for the AlphaBit Club, main goal was to attract companies to finance the club events.
- Technologies used: React, Tailwind CSS, Headless UI, Firebase, Axios.

Innovium 03/2023 - 06/2023
github.com/boite-informatique/startup-front

- A web application to digitize the application process for startups and final-year projects, developed as part of my 3rd year in a team.
- Technologies used: Svelte TypeScript, Tailwind CSS, DaisyUI, svelte-i18n.

ESI Hub 03/2022 - 06/2022
github.com/boite-informatique/esi-hub-backend

- Contributed to the backend development of a social network platform for students at ESI SBA, developed as part of my 2nd year in a team.
- Technologies used: Node.js, Express, MongoDB, Mongoose, JWT, Multer, Socket.io.

Skills

Languages: C, C++, Python, R, Java, JavaScript.

Web Development: Svelte, React, Node.js & Express.js, REST APIs & GraphQL.

Networking: IoT, Routing Protocols (RPL, EIGREP, OSPF, BGP).

AI & ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.

Mathematics: Numerical Linear Algebra, Probability & Statistics, Numerical Methods, Graph Theory.

IoT & Robotics: Raspberry Pi, Arduino, ROS, Motion & Trajectory Planning, Collision Avoidance (ORCA).

Tools: Git & GitHub, Docker, Contiki-NG/Cooja, GNS3, NS-3, LaTeX.

Languages

English	French	Arabic	German
Fluent - IELTS 7.5 (C1)	Fluent - TCF Certificate C1	Native language	Goethe Certificate B1

Volunteering

Alphabit Club - Technical Team Member — 2022 - 2024
Member of the technical team responsible for creating and maintaining the Alphabit website, authoring hackathon challenges, and performing other technical tasks during events. — Algeria
alphabit.club

Internationale Jugendgemeinschaftsdienste (ijgd), Germany — 02/2024
Completed ijgd training on youth work and group leadership, allowing me to lead international workcamps in Germany and other EU countries. — Hannover, Germany
www.ijgd.de

EYE 2025 (European Youth Event) — 06/2025
Participation in the European Youth Event organized by European Parliament. — Strasbourg, France

Jeunesse Volontaire — 2016 - 2024
A local environmental and social action organization in Algeria. — Sidi Bel Abbes, Algeria

IJGD (08/2023 - 09/2023) — 09/2023
A volunteer workcamp in Germany, to assist with the showcasing of art exhibitions. — Neunkirchen, Germany

IJGD (08/2019 - 09/2019) — 09/2019
A volunteer workcamp in Bärnau-Tachov, for the maintenance of a local historical park. — Bärnau, Germany

IJGD - Camp for Future — 08/2018
A volunteer workcamp focused on anti-coal activism, organized at the Hambach Forest. — Hambach, Germany

Referees

Dr. M. Boudouaia
Researcher and Lecturer at CESI LINEACT (France)
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ResearchGate

Dr. A. Rahmoun
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Google Scholar

Dr. F. Bendaoud
Associate Professor and Head of the AI Department, ESI-SBA (Algeria)
Email: f.bendaoud@esi-sba.dz
Google Scholar

Interests

Cybersecurity, Internet Freedom, FOSS (Free Open Source Software), Web3, Football, Geography, Urban planning, Travel and discovering other cultures.